Akshit Singh

Reverse Engineer | Mobile Vulnerability Researcher

+91 9868520459 | akshitdavps@gmail.com | https://www.linkedin.com/in/akshit-singh-69ba5a192/
https://github.com/it4ch1-007 | https://it4ch1-007.github.io/about | https://x.com/akshit_it4ch1

ABOUT

IIT Roorkee graduate and Former member of InfoSecIITR specializing in Mobile Vulnerability Research, Reverse Engineering, Malware Analysis, and Exploit Development. Proven expertise in low-level OS internals, vulnerability discovery, and offensive tooling. Skilled in bypassing modern exploit mitigations, binary analysis, and weaponizing vulnerabilities across mobile ecosystems

SKILLS

Soft Skills: Time Management, Communication, Problem-solving, Teamwork, Leadership, Critical Thinking

Programming Languages: Java, Kotlin, C/C++, Objective-C, Python, Rust, Assembly - ARM/x86, SQL, NodeJS, Shell scripting

Software Packages: Frida, IDA Pro, Ghidra, Theos, Hopper Disassembler, Jadx, Sandhook, Apktool, ADB, Android Studio, Xcode, Wireshark, Objection, Reverse Engineering automation, Magisk, Dopamine, AFL++ Fuzzer

DevOps: Firebase, Play Integrity API, Linux, Docker, Git, Kubernetes, Network security, Database security, AWS, GCP, Azure

Cybersecurity: Car CAN Reverse Engineering, AI Security and Red Teaming, Android Internals, iOS internals, Android/iOS Debugging, Fuzzing, eBPF, Blockchain security, Kernel exploitation (Windows/Linux), Windows internals, Exploit Mitigation Bypasses (PAC, MTE, ASLR), Malware Internals, Windows kernel development, Database security, Linux kernel development, Game security, IoT/Hardware security, Heap exploitation

EXPERIENCE

Android/iOS Reverse Engineer | Talsec | https://www.talsec.app

Oct 2024 - Present

- Conducted deep-dive architectural research into Android and iOS OS internals to **analyze advanced persistence**, jailbreaking, and **instrumentation techniques (Magisk, Dopamine, palera1n)** alongwith malicious techniques used by **mobile malwares**.
- Reverse engineered banking applications and security modules like **ICICI banking mobile app, Yono SBI App, BOSCHK, Lookout, Promon and Lockin - for Shamiko Magisk, Zygisk hiders, Nox , Bluestacks emulator, Dopamine, palera1n and Hook detection.**
- **Engineered native C++ modules that interact directly with process memory** to detect runtime hooking tools like FRIDA and Objection.
- Developed **Anti-root, Anti-hooking, Device binding, Anti-multiinstancing and Anti-emulator methods** for **freeRASP**.
- **Skills: FRIDA, Magisk, C/C++, Kotlin, Android/iOS internals,Vulnerability Discovery, Exploit Mitigation Analysis, and Jailbreak/Rootkit Internals.**

Cloud Security Researcher | CloudDefense.AI | https://www.clouddefense.ai/

July 2024 - Oct 2024

- **Engineered custom security tooling in Golang and Rust to map attack surfaces** and analyze configurations across enterprise cloud environments (AWS, GCP, Azure) and manage databases (AlloyDB, DynamoDB, MongoDB Atlas).
- Performed **vulnerability scanning** to secure **Personally Identifiable Information** (PII) within databases including malware detection and **attack path analysis** by deploying cloud security tools and developing **Cloud SDKs for GoFiber using Golang**.
- Executed **advanced privilege escalation** and **account takeover techniques against cloud platforms** and user accounts and **engineered offensive SDKs in Golang** to exploit backend infrastructure, using lateral movement and unauthorized access to database systems.
- **Skills: Golang, Rust, C/C++, AWS, GCP, MongoDB Atlas, Java**

Offensive Android Security Researcher | Forensics Cybertech | Ministry of Home Affairs India

Mar 2024 - May 2024

- Conducted offensive security research for the **Ministry of Home Affairs (India)**, analyzing and developing Proof-of-Concepts for 100+ **Android kernel and application-layer vulnerabilities** to achieve unauthenticated, system-level privileges.
- **Reverse-engineered 50+ complex N-day vulnerabilities** using IDA Pro, Ghidra, and pwndbg to perform Root Cause Analysis, **bypass modern OS mitigations**, and **reconstruct reliable exploit chains.**
- Implemented low-level kernel bugs by engineering **custom C-based exploits**, focusing on **Local Privilege Escalation** and memory corruption primitives across **major OEM firmwares (Xiaomi, Samsung, Apple**).
- Leveraged advanced debugging (GDB) and emulation frameworks to simulate exploit environments, successfully validating root-level access and demonstrating post-exploitation data extraction techniques on target devices.
- **Skills: IDA Pro, adb debugging, Frida, C/C++, Python, Android Studio, GDB debugging**

EDUCATION

Bachelors of Technology, IIT Roorkee | 2022 - 26
Majors: Electrical Engineering

Bachelors of Technology, NSUT Delhi | 2021 - 22
Majors: Computer Science Engineering

DAV Public School, New Delhi
Completed High School Education

CERTIFICATIONS

Awiros AppaThon , IIT Delhi
Developed a mobile application based on OpenCV for traffic monitoring using Awiros automated App development framework

3rd globally in PragyanCTF 2024, NIT Trichy

3rd globally in CodefestCTF2024, IIT BHU

4th globally in IncognitoCTF2024

ACHIEVEMENTS

- Contributor at **the FRIDA project**
- Presented on FRIDA internals at **Mobile Security Conference 2025 at Prague, Czech Republic**
- Qualified for **DCTF finals at DefCamp Conference in Romania**
- **Qualified for Awiros AppAThon finals at IIT Delhi**
- 1st in **CSAW'Quals 2025** (India region) organized by **New York University of Technology**
- 1st globally in **VishwaCTF 2024**
- 1st globally in **JerseyCTF 2024,** organized by the **New Jersey Institute of Technology**
- Qualified for the final round of **TrustLabs CTF held at IIT Bombay** among more than **10 lakh national players.**
- Qualified for **HackDay** CTF Finals 2024, **held at ESIEE, Paris**
- 6th globally in **b0il3rsCTF 2024**
- 9th globally in **BYUCTF 2024**
- 3rd globally in PragyanCTF 2024, **organized by NIT Trichy**
- 5th globally in **UMASSCTF 2024**
- 3rd globally in CodefestCTF 2024, **organized by IIT BHU**
- Won Times of India **National English Debate Contest.**
- Finished **7th** at **EY DSCI CTF,2024**

PROJECTS

SilentPulse | **SeLinux Bypass, Magisk's libSu, SandHook implementation**

https://github.com/it4ch1-007/audioRecorder

- An Android application that gains root inside the victim's device using **Magisk's libSu API** and uses a background service to inject a malicious hooking library into the processes' memory and **modify the process' memory** without any notification on the victim's screen.
- It **bypasses the SELinux** policies using the **supolicy** tool and a **custom Sandhook mechanism** to hook into the audioserver process of the device, detect when a call is made by or to the device, and collect the **PCM data packets** of VoIP calls made through the device.
- It **bypasses any notification prompts** to the user to execute all its functionalities stealthily in the background of the device.
- It uses **priority queues for real-time storage and reordering** of outgoing and incoming packets in files inside external storage without explicit permissions using **FFT algorithm** to determine amplitude of PCM packets in real-time.
- Uses **Android Debug Bridge, AIDL interface, C/C++, Java, Kotlin, SuPolicy, SandHook, x86_64 assembly**

Overlay Hacker | **Shizuku, ADB Bridge, Android native programming, AIDL interfacing**

https://github.com/it4ch1-007/AttackerAppJava

- An Android **pentesting** application that uses **Shizuku API** and the **IShizuku** service to elevate privileges **without jailbreaking** the device.
- Uses **Overlay attacks, Multithreading and API calls** to simulate an environment that **steals the users' credentials** without any permission prompt and sends them to the attacker's system as well as stores them inside privileged environment.
- Leveraged Shizuku and **system APIs** to perform privileged operations **accessing restricted system resources.**
- Able to execute any ADB commands remotely using Wi-Fi debugging bridge of the device and elevating privileges to ADB permissions.
- Uses **ADB Bridge, AIDL interface, C/C++, Java, Kotlin, Android Studio, JADX, Shizuku, SELinux bypass**

Modder.exe | **C++, Windows API, Assembly Programming**

https://github.com/it4ch1-007/Modder_exe

- A C++-based Windows game reversing tool that includes features like **DLL hooking, Deobfuscation, Memory scanning**, **Disassembling, Debugging**, **Call logging**, **Windows Defender evasion** and **Chat proxy using packet capturing**.
- Implemented automated **instruction pattern matching** and **runtime instrumentation** to speed up function discovery, game artifact discovery, code-flow analysis, and **dynamic behavioral analysis** of native game binaries.
- Capable of **logging 100 packets per second** and their **dynamic inspection** using Windows APIs, Assembly language, efficient and optimized C++ functions, user-friendly CLI console for ease of use and efficiency.
- **Uses C++, Assembly, PowerShell, Batch, Windows API, Windows Internals, Unicorn Engine, Capstone Engine, Windows AV Evasion.**

Cobra | **Rust, Windows Internals, Malware Analysis, Powershell, Batch, AV Evasion**

https://github.com/it4ch1-007/Cobra

- Developed an **advanced, memory-safe Command and Control** in Rust, achieving a undetectable status against 25+ modern **Antivirus and EDR solutions** through **custom cryptographic payloads** and **Windows AV Evasion techniques**.
- Developed not to trigger any AV checks and exploit the basic mechanisms of the **Windows subsystem with modern persistence techniques, Credential stealing techniques, Lateral Movement techniques and unauthorized signature forging techniques.**
- Demonstrated deep expertise in **OS architecture and memory manipulation** by developing **advanced injection and evasion primitives**, including **APC Queue injection, dynamic API hashing, DLL/Code injection, and Process Environment Block** traversal for anti-debugging
- Uses **Powershell and Batch scripting to bypass UAC checks** and implement new AV evasion techniques like **Custom crypto algorithms**, **PEB traversal** debug checks, **WLAN Parsing**, APC Queue injection, Code injection, DLL injection, API hashing and more.

EXTRACURRICULAR

Reverse Engineer | **InfoSecIITR**

https://infoseciitr.in/

- Represented an elite, **globally recognized CTF team InfoSecIITR**, competing in high-stakes international **Capture The Flag and vulnerability discovery competitions.**
- Specialized in r**everse engineering and binary analysis**, analyzing obfuscated code, and **developing exploits** during live global hackathons.
- Architected **complex offensive security challenges** (Reverse Engineering & Pwn) for international events like **BackdoorCTF and n00bCTF,** simulating real-world memory corruption, binary exploitation, and mitigation bypass scenarios.
- Audited and **secured the underlying CTF platform infrastructure** while actively mentoring campus talent in cybersecurity.

Organizer, BackdoorCTF | **2024** | **2025**

https://ctftime.org/ctf/49/

- Architected **advanced Reverse Engineering challenges** for BackdoorCTF—a premier global cybersecurity competition hosting 500+ elite teams; simulating **real-world binary obfuscation and exploitation scenarios.**
- Audited and hardened the **underlying GCP-hosted CTFd infrastructure**, ensuring high availability and resilience against active exploitation during the global event.

Organizer, ScytheCTF | **2024**

https://www.cognizance.org.in/events/centerstage/27/61

- Developed **Reverse Engineering and Binary Exploitation challenges for ScytheCTF (Cognizance IIT Roorkee), a national-level offensive security competition hosting 100+ participants.**
- Audited and secured the CTFd platform infrastructure while directing on-site technical operations, serving as the **technical lead for challenges and exploit validation.**

PUBLICATIONS

- *POC CVE-2020-0022*
- *POC CVE-2020-0096*
- *Android Malware Analysis*
- *Shizuku Article*
- *Android Signature Model and APKSignatureKiller*
- *Frida Internal Working*
- *AutoBot_exe Analysis*
- *IDA Pro internals*
- *Game Reversing Series*